Domain Name System

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Content

- Domain Name System (DNS)
  - Characterization
  - Organization
  - Configuration
  - Commands, Primitives
  - Internationalized Domain Name (IDN)

DNS

- IP addresses (e.g., 85.122.23.145, 2001:0db8:0001:0000:0000:0ab9:C0A8:0102) are difficult to remember
- A **domain name system** is used to map IP addresses to domain names and vice versa
- Domain names are organized in hierarchies

- DNS is defined in many RFC documents (e.g., 1034, 1035, 1123, 2181, 5890, 5891, etc.)
- www.statdns.com/rfc/

DNS| organization

- Initial: **/etc/hosts** – pairs (name, IP)
  - Scalability problems

- Actual: DNS consists of a hierarchical naming scheme and a distributed database system to implement this naming scheme

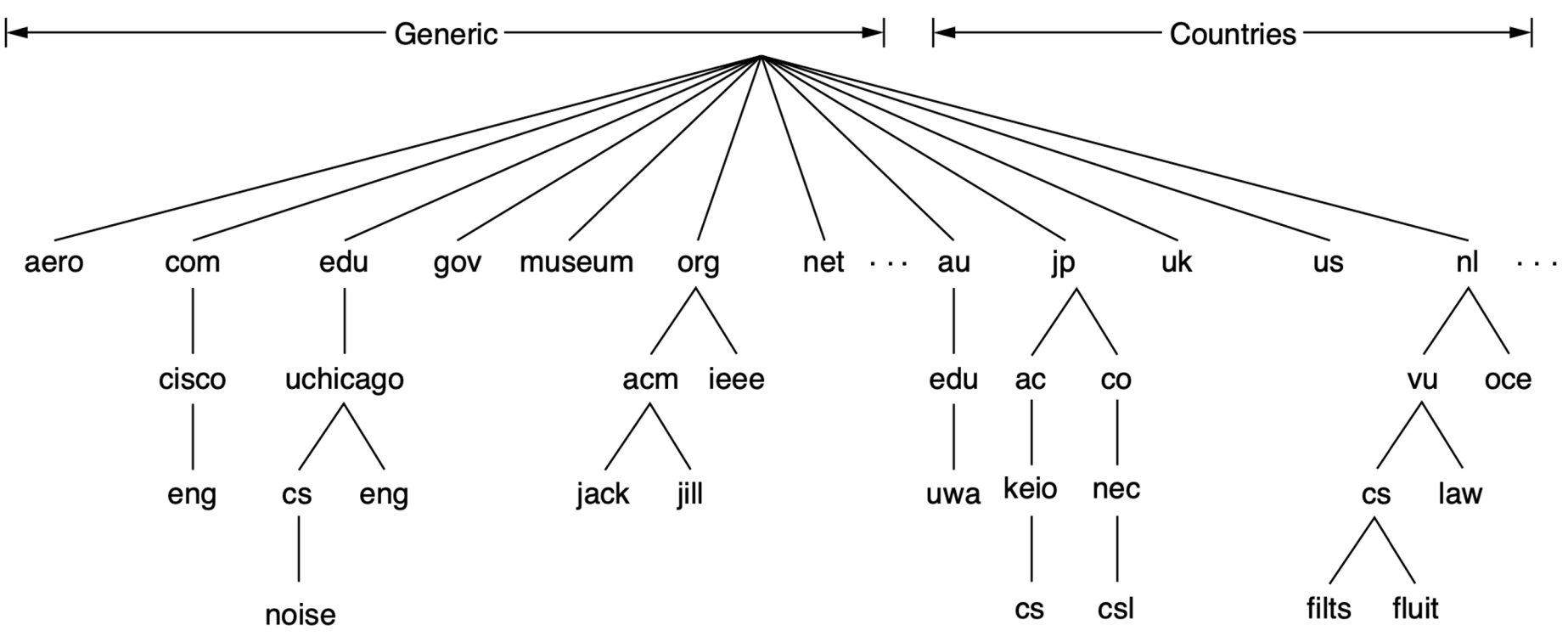


Figure. A portion of the domain name space on the Internet

[Computer Networks, 2021, Andrew S. Tanenbaum et. al.]

DNS| domain types

- **Primary** (Top Level Domains – TLD)
  - For Internet Infrastructure – one domain: **.arpa** ARPA (Address and Routing Parameter Area)
    - administered by IANA
    - was the first TLD
    - initially used for the transition to the DNS
    - currently used in a few particular contexts (e.g., reverse DNS resolution)
  - State (*ccTLD*) – states code: .ro, .fr, .jp, … (as defined in ISO 3166)
  - Generics: .biz, .com, .info, .name, .net, .org, .pro, …
    - in 2011 there were only 22 gTLDs
    - since 2011, ICANN allowed the registration of custom gTLDs (the initial cost for applying was approximately 200000 USD)

DNS| domain types

- **Primary** (Top Level Domains – TLD)
  - Sponsored: .aero, .edu, .gov, .int, .jobs, .mil, .tel, …
  - Reserved: .example, .invalid, .localhost, .test, … (RFC 2606)
    - never exposed in the Internet
    - to be used for local testing, in documentations, etc.
    - in July 2024, ICANN approved a new TLD, .internal
      - reserved for private use within internal networks (similar to certain classes of local IP addresses)
      - it will not be delegated to the DNS root
      - the purpose is to standardize the use of a TLD within internal networks, so that different organizations no longer create their own ad-hoc TLDs, as they have done so far
      - https://www.icann.org/en/board-activities-and-meetings/materials/approved-resolutions-special-meeting-of-the-icann-board-29-07-2024-en#section2.a

DNS| domain types

- **Primary** (Top Level Domains – TLD)
  - IDN TLDs (Internationalized Top-Level Domains)

    http://example.test -> http://παράδειγμα.δοκιμή (Greek) (experimental)

    http://example.test -> http://例え.テスト (Japanese) (experimental)

    http://example.test -> http://उदाहरण.परीक्षा (Hindi) (experimental)

    .中国 (ccTLD for China)

    .닷넷 , .닷컴 (gTLDs in South Korea, meaning “dotnet” and “dotcom”)

  - Pseudo-domains: .bitnet, .local, .root, .uucp, .onion, .crypto, .bitcoin, .eth …
    - not in the world wide official DNS

DNS| domain types

https://www.iana.org/domains/root/db

Domain Names

- Overview
- **Root Zone Management**
  - Overview
  - **Root Database**
  - Hint and Zone Files
  - Change Requests
  - Instructions & Guides
  - Root Servers
- .INT Registry
- .ARPA Registry
- IDN Practices Repository
- Root Key Signing Key (DNSSEC)
- Reserved Domains

Root Zone Database

The Root Zone Database represents the delegation details of top-level domains, including gTLDs such as .com, and country-code TLDs such as .uk. As the manager of the DNS root zone, we are responsible for coordinating these delegations in accordance with our policies and procedures.

Much of this data is also available via the WHOIS protocol at whois.iana.org.

DOMAIN	TYPE	TLD MANAGER
.aaa	generic	American Automobile Association, Inc.
.aarp	generic	AARP
.abarth	generic	Fiat Chrysler Automobiles N.V.
.abb	generic	ABB Ltd
.abbott	generic	Abbott Laboratories, Inc.
.abbvie	generic	AbbVie Inc.
.abc	generic	Disney Enterprises, Inc.
.able	generic	Able Inc.
.abogado	generic	Minds + Machines Group Limited
.abudhabi	generic	Abu Dhabi Systems and Information Centre
.ac	country-code	Network Information Center (AC Domain Registry) c/o Cable and Wireless (Ascension Island)
.academy	generic	Binky Moon, LLC
.accenture	generic	Accenture plc

DNS| domain types

- Domain name
  - Sub-tree of the domain tree
  - The physical topology is not taken into consideration

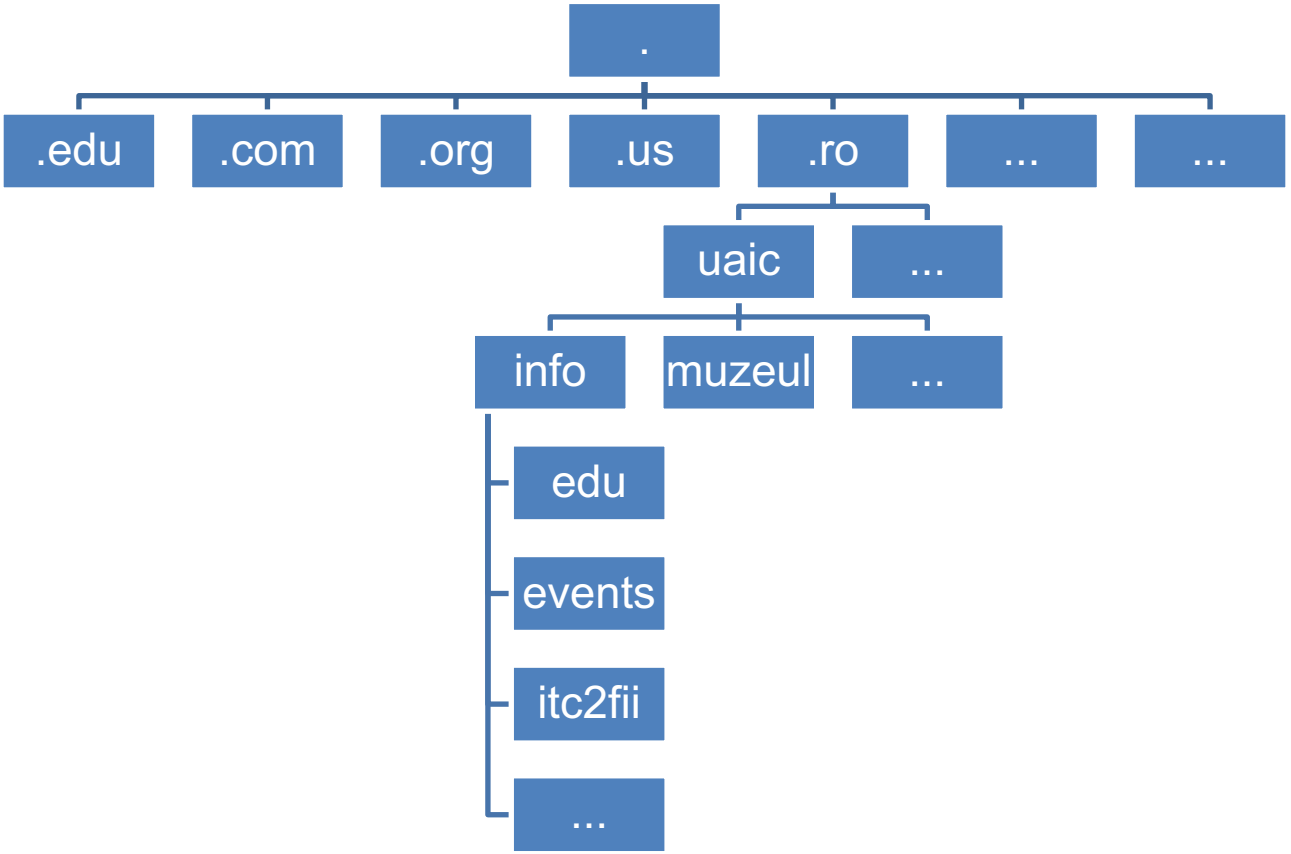
- **Sub-domains**
- **Name of computers** (hosts)

! full path names cannot exceed 255 characters

! component names cannot exceed 63 characters

DNS

- Example:



DNS| organization

- Rules to allocate domain names:
  - Each domain controls how its subdomains are assigned
  - To create a new subdomain, permissions are requested from the upper domain (each domain will have an authority at a certain level)
  - Domain name assignment respects organizational boundaries, not those of the networks
  - A certain level of hierarchy can be controlled by multiple servers

DNS| organization

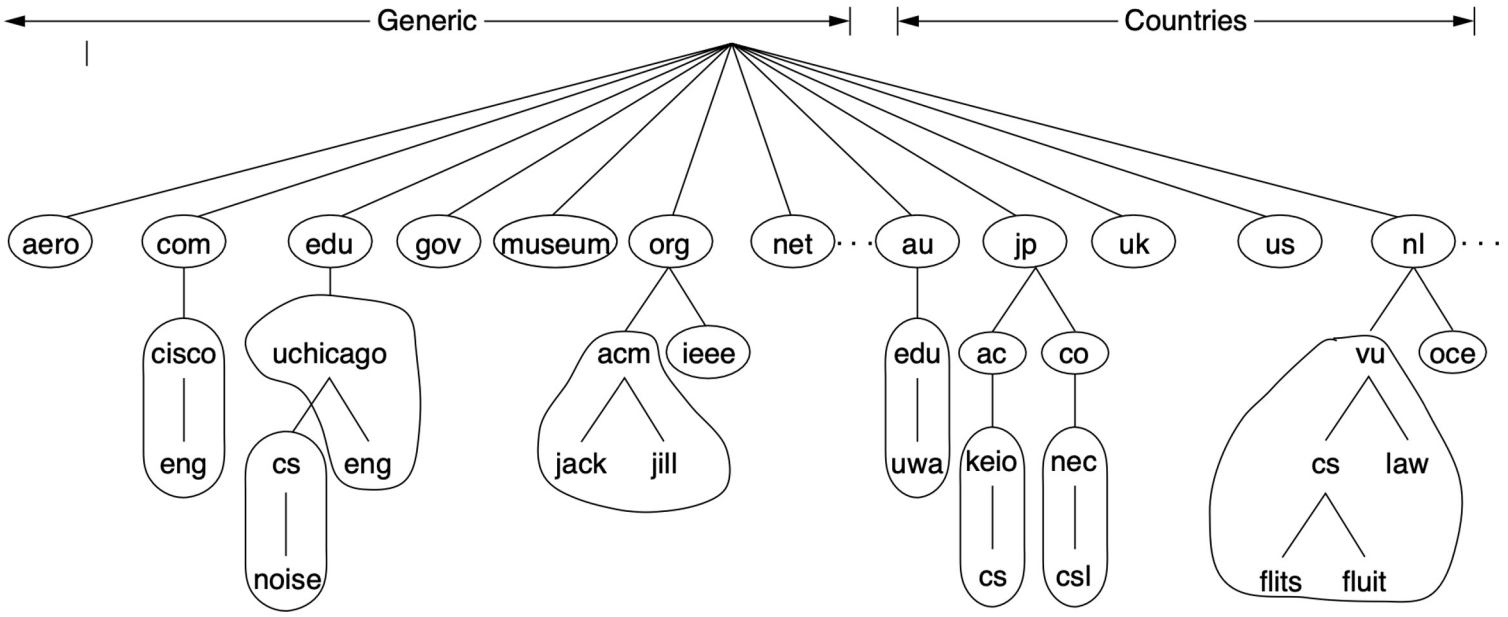
- Name servers
  - Theoretically, a single name server can contain the entire DNS database and can respond to all requests
  - Problems: loading and *"single point of failure"*

  - DNS name space is divided into non-overlapping areas

DNS| organization

- Name servers

  Example: A possible division of DNS namespace in areas



[Computer Networks, 2021, Andrew S. Tanenbaum et. al.]

DNS| organization

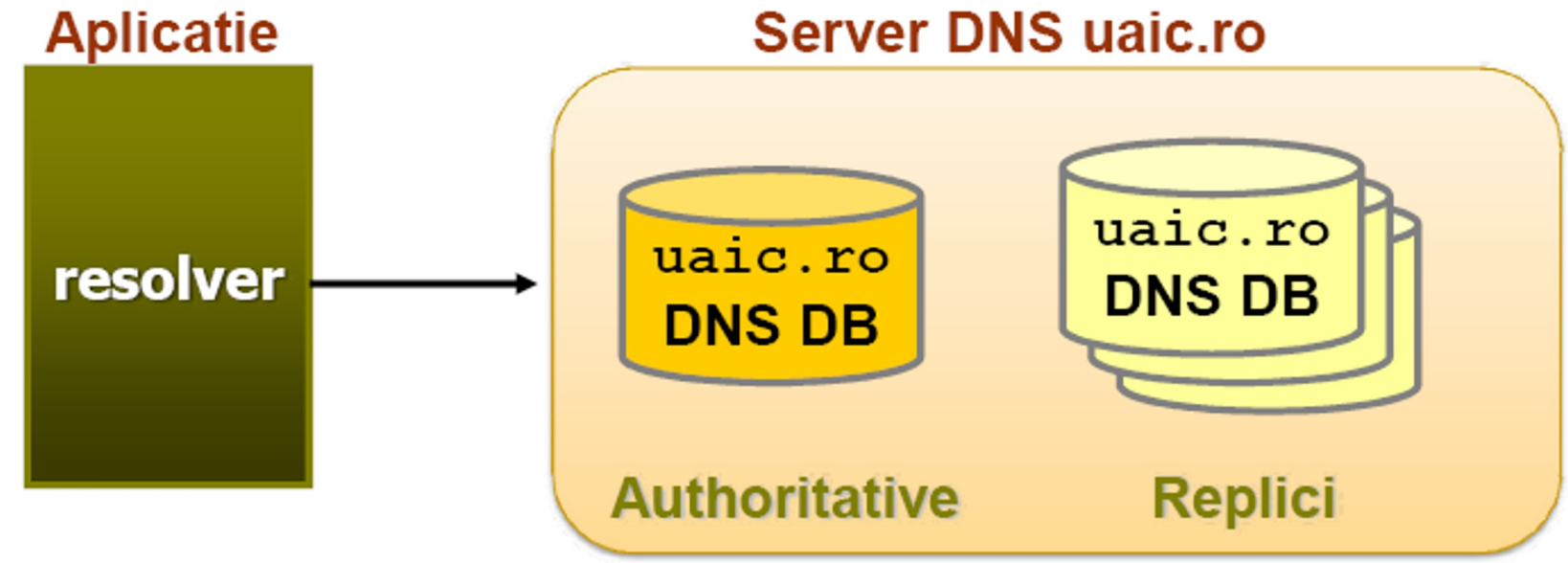
- Name servers
  - There is a *primary/authoritative name server* that manages a certain domain, and possibly, more secondary servers that contain replicated databases
  - TCP is used for DNS replication
  - UDP is used for queries (lookups)
    - including in the new standard DNS-over-QUIC (DoQ) (RFC 9250)

! Nowadays, also TCP is used for queries, for security reasons (see DNSSEC (RFC 9364), DNS-over-TLS (DoT) (RFC 7858), DNS-over-HTTPS (DoH) (RFC 8484))
blog.apnic.net/2022/09/02/doh-dot-and-plain-old-dns/

stats.labs.apnic.net/edns

DNS| organization

- DNS Client
  - Called ***resolver***, sends an UDP packet to a DNS server; the server seeks the name and returns the IP address



[Retele de calculatoare – curs 2007-2008, Sabin Buraga]

DNS| organization

- Example of name server implementations: **BIND** (Berkeley Internet Name Domain), CoreDNS, Microsoft DNS, PowerDNS, etc.

- As interactive resolvers (clients), the following commands can be used: **nslookup**, **host**, and **dig**.

DNS| queries

- Queries:
  - **Recursive** – if a DNS server does not know the address for the requested name, then it will query another DNS server
  - **Incremental** – if the DNS server does not know how to respond, it will return an error and another DNS server address (also called *referral*) that may know the answer to the query

learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2003/cc775637(v=ws.10)

DNS| queries

- Each domain is associate with a set of resource records (***resource record*** – **RR**)
- The mechanism:
  - The request: the *resolver* sends a domain name
  - The response: the resource records associated with that name (stored in DNS database)

DNS creates a correspondence between the domain names and the resource records

DNS| queries

- RR general form:

  Domain_Name Time_to_live Class Type Value

  Domain name – specifies the field covered by this record

  Time_to_live – gives an indication of how stable the record is

  Class – always has the value *IN* (for Internet data)

  Type – specifies the registration type

  Value – this field can be a number, a domain name, or an ASCII string; the semantics depend on registration

DNS| queries

Table. Main DNS resource records

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

[Computer Networks, 2021, Andrew S. Tanenbaum et. al.]

DNS| queries

Type - specifies the registration type

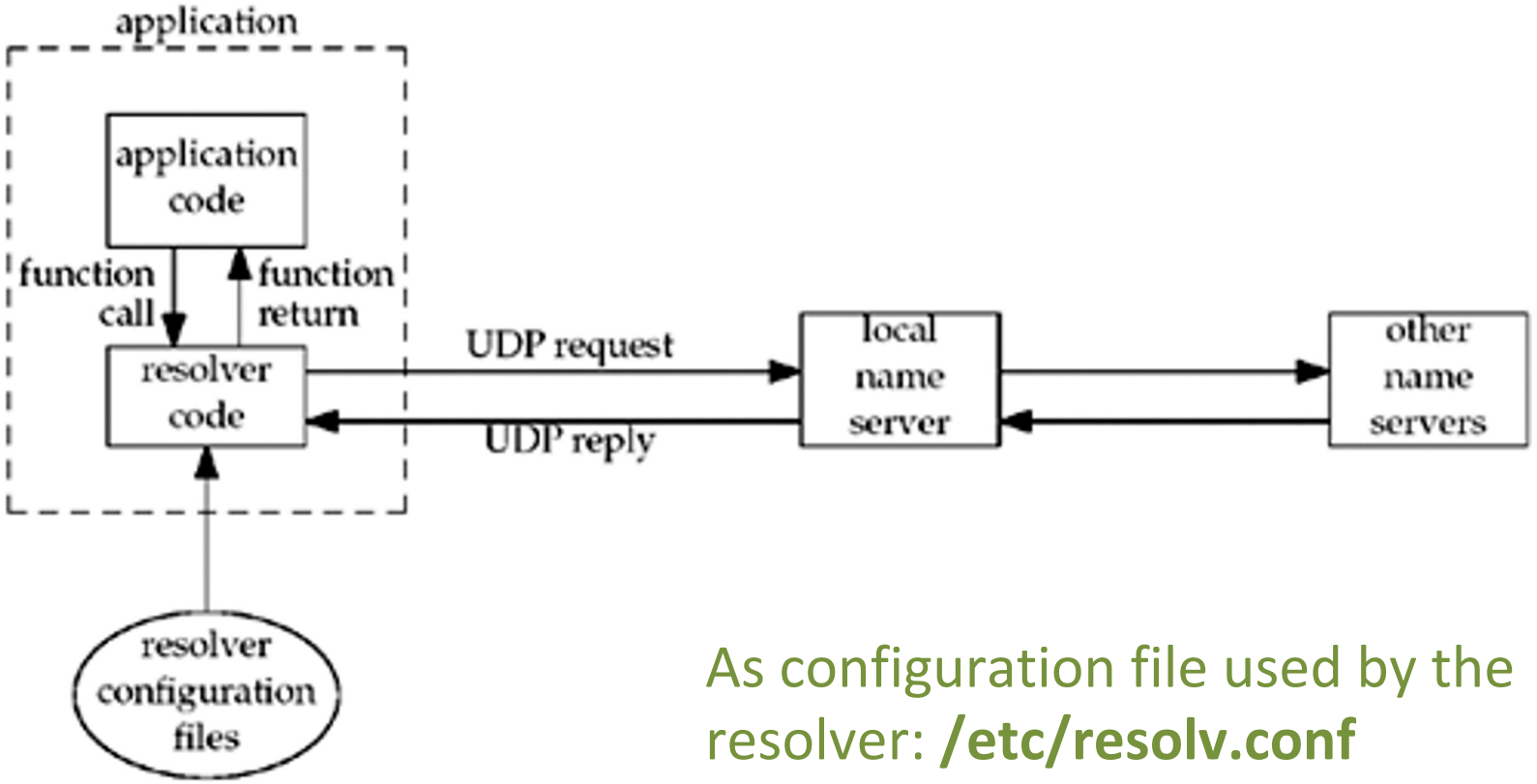
- **SOA** (start of authority) – the current domain, administrator e-mail address, etc.
- **A** – host IP (for IPv4)
- **AAAA** – host IP (for IPv6)
- **MX** (mail exchangers) – specifies the domain name ready to accept mail for the specified domain
- **CNAME** (*canonical name*) – allows creation of pseudonyms
- **PTR** (pointer) – Alias for IP address
- **TXT** – uninterpreted text (comments)

DNS| configuration

- Example of a file containing a DNS zone specification

```
; Zone file for axiologic.ro
;
; The full zone file
;
$TTL 3D
@       IN      SOA     ns1.axiologic.ro. abss.axiologic.ro. (
                        2007050103        ; serial, todays date + todays serial #
                        14400             ; refresh, seconds
                        7200              ; retry, seconds
                        1209600           ; expire, seconds
                        1D )              ; minimum, seconds
;
@                NS      ns1.axiologic.ro.              ; Inet Address of name server
@                NS      ns2.axiologic.ro.              ; Inet Address of name server
@                MX      5 mailx.axiologic.ro.     ; Primary Mail Exchanger
;
localhost        A       127.0.0.1
axiologic.ro.    A       72.249.105.153
www              A       72.249.105.153
mailx            CNAME   axiologic.net.
mail             A       207.210.101.144
ftp              A       72.249.105.153
axiologic.ro. IN TXT "v=spf1 mx mx:mailx.axiologic.ro. ~all"
ns1              A       207.210.101.144
ns2              A       207.210.101.216
~
(END)
```

DNS| clients, resolvers, servers



As configuration file used by the resolver: **/etc/resolv.conf**

[Unix Network Programming, 2003, R. Stevens B. Fenner, A. Rudoff]

DNS| configuration

- /etc/resolv.conf file:

```
[adria@thor ~] $ cat /etc/resolv.conf
domain info.uaic.ro
search info.uaic.ro
nameserver 85.122.16.1
nameserver 85.122.16.4
[adria@thor ~] $
```

DNS| reverse queries

- Problem:
  - If we have an address, which will be its symbolic name? (*reverse DNS resolution* or *reverse DNS lookup*)

Example:

1)

```
[adria@ns1 ~]$ host 85.122.23.1
1.23.122.85.in-addr.arpa domain name pointer thor.info.uaic.ro.
[adria@ns1 ~]$
```

2) 2001:db8::567:89ab

b.a.9.8.7.6.5.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.8.b.d.0.1.0.0.2.ip6.arpa

DNS| optimizations

Spatial proximity: local servers will be queried more often than others at the distance

Temporal proximity: if a set of fields are referenced repeatedly, then the DNS caching mechanism is used

For each DNS entry a TTL (*time to live*) value is set

Replication is also used (multiple servers, multiple root servers) – the nearest (geographically) server will be interrogated

DNS| commands

As interactive resolvers, we can use:

- **nslookup**
- **dig**
- **host**
- **whois**
- …

DNS| nslookup

Usage examples:

$ nslookup www.info.uaic.ro

- Returns a RR of type A, using the local DNS server

```
-sh-4.2$ nslookup www.info.uaic.ro
Server:         85.122.16.1
Address:        85.122.16.1#53

Name:   www.info.uaic.ro
Address: 85.122.23.162
```

Host Lookup

$ nslookup 85.122.23.1

- Returns a RR of type PTR for 85.122.23.1 in *in-addr.arpa* domain hierarchy

```
-sh-4.2$ nslookup 85.122.23.1
1.23.122.85.in-addr.arpa        name = mail.info.uaic.ro.
1.23.122.85.in-addr.arpa        name = thor.info.uaic.ro.
```

Reverse IP Lookup

[http://www.zytrax.com/books/dns/ch3/]

DNS| nslookup

Usage examples:

$ nslookup www.axiologic.ro [server]

- returns a RR of type A using a specified DNS server

```
-sh-4.2$ nslookup www.axiologic.ro 13.248.158.159
Server:         13.248.158.159
Address:        13.248.158.159#53

Name:   www.axiologic.ro
Address: 185.53.177.53
```

Host Lookup

$ man nslookup

DNS| dig

dig – a tool more powerful than nslookup

Usage example:

$ dig www.info.uaic.ro A

```
-sh-4.2$ dig www.info.uaic.ro A

; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.el7_9.14 <<>> www.info.uaic.ro A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 7888
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 3, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.info.uaic.ro.              IN      A

;; ANSWER SECTION:
www.info.uaic.ro.       86400   IN      A       85.122.23.162

;; AUTHORITY SECTION:
info.uaic.ro.           86400   IN      NS      onix.uaic.ro.
info.uaic.ro.           86400   IN      NS      ns.iasi.roedu.net.
info.uaic.ro.           86400   IN      NS      orion.uaic.ro.

;; ADDITIONAL SECTION:
ns.iasi.roedu.net.      86400   IN      A       192.129.4.100
ns.iasi.roedu.net.      86400   IN      AAAA    2001:b30:1:100::100
onix.uaic.ro.           600     IN      A       85.122.16.4
orion.uaic.ro.          600     IN      A       85.122.16.1

;; Query time: 0 msec
;; SERVER: 85.122.16.1#53(85.122.16.1)
;; WHEN: Fri Nov 24 22:27:07 EET 2023
;; MSG SIZE  rcvd: 207
```

DNS| host

host

Usage example:

```
-sh-4.2$ host 128.30.52.45
45.52.30.128.in-addr.arpa domain name pointer blackhole.w3.org.
```

DNS| whois

$ whois ibm.com

```
Domain Name: ibm.com
Registry Domain ID: 1555443_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.corporatedomains.com
Registrar URL: www.cscprotectsbrands.com
Updated Date: 2023-03-16T01:11:31Z
Creation Date: 1986-03-19T00:00:00Z
Registrar Registration Expiration Date: 2024-03-20T04:00:00Z
Registrar: CSC CORPORATE DOMAINS, INC.
Sponsoring Registrar IANA ID: 299
Registrar Abuse Contact Email: domainabuse@cscglobal.com
Registrar Abuse Contact Phone: +1.8887802723
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Domain Status: serverDeleteProhibited http://www.icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited http://www.icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited http://www.icann.org/epp#serverUpdateProhibited
Registry Registrant ID:
Registrant Name: IBM DNS Admin
Registrant Organization: International Business Machines Corporation
Registrant Street: New Orchard Road
Registrant City: Armonk
Tech Name: IBM Corporation
Tech Organization: International Business Machines (IBM)
Tech Street: New Orchard Road
Tech City: Armonk
Tech State/Province: NY
Tech Postal Code: 10598
Tech Country: US
Tech Phone: +1.9192544441
Tech Phone Ext:
Tech Fax: +1.9147654370
Tech Fax Ext:
Tech Email: dnstech@us.ibm.com
Name Server: ns1-99.akam.net
Name Server: usc3.akam.net
Name Server: usc2.akam.net
Name Server: eur2.akam.net
Name Server: ns1-206.akam.net
Name Server: asia3.akam.net
Name Server: usw2.akam.net
Name Server: eur5.akam.net
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2023-03-16T01:11:31Z <<<
```

DNS| primitives

- We do not have to develop a resolver in order to find out the IP address of a host
- Main functions:
  - **gethostbyname(); gethostbyaddr() -> obsolete**
  - **getaddrinfo(); getnameinfo() -> recommended**
- additional:
  - **getservbyname(); getservbyport()**

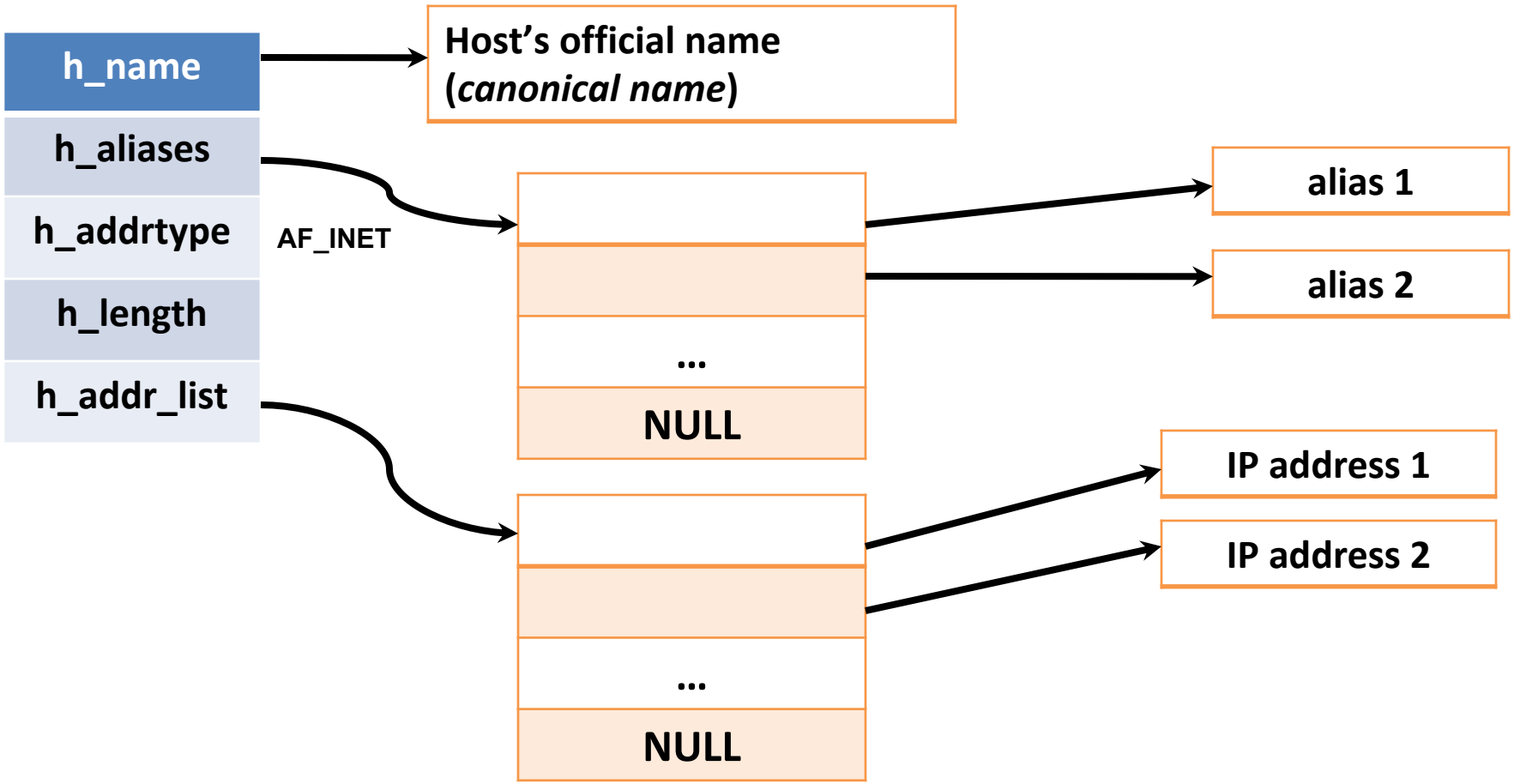
DNS| primitives

One of the used structures: **hostent**

```
struct hostent {
    char *h_name;       /* official name (canonical) */
    char **h_aliases;   /* aliases */
    int h_addrtype;     /* AF_INET */
    int h_length;       /* address length: 4 or 6 */
    char **h_addr_list; /*pointers to IP addresses */
};
```

DNS| primitives

hostent structure:



DNS| gethostbyname()

obsolete

```
#include <netdb.h>
struct hostent *gethostbyname
                    (const char *hostname);
```

- In DNS terms, gethostbyname() does a request for a type **A** record
- Obs.: gethostbyname() is used mostly for IPv4

DNS| gethostbyname()

obsolete

- Returns:
  - On success, it returns a pointer to **hostent**, which contains the host's IP address
  - On error, it returns NULL, and h_errno indicates the error number:
    - HOST_NOT_FOUND
    - …
    - NO_RECOVERY
    - …

Constants defined in netdb.h

DNS| gethostbyname()

obsolete

- Usage example: setting a symbolic name instead of an IP address in a *sockaddr_in* structure:

```
struct sockaddr_in server;
struct hostent *hos;
if(!( hos = gethostbyname("fenrir.info.uaic.ro") )
  {/*Error on resolving address*/}
server.sin_family=AF_INET;
  /* we take the IP address from hos structure */
memcpy(&server.sin_addr.s_addr, hos->h_addr_list[0],
                        sizeof(hos->h_addr_list));
server.sin_port=htons(4321);
```

DNS| gethostbyaddr()

obsolete

```
#include <netdb.h>
        struct hostent *gethostbyaddr (
                        const char *addr,
                        socklen_t len,
                        int family);
```

- In DNS terms, **gethostbyaddr()** does a request to the nameserver for a **PTR** record in *in-addr.arpa* domain
- Return value: On success, it returns a pointer to **hostent**, which contains the official name of the *host*; On error, it returns NULL, and h_errno variable indicates the error

Obs.: gethostbyaddr() is used mostly for IPv4

DNS| getservbyname()

```
#include <netdb.h>
struct servent *getservbyname (const char *servname, const char
    *protoname);
```

- Return value: On success, a pointer to *struct servent*; On error, NULL

```
        struct servent {
            char *s_name;      /* official name of the service*/
            char **s_aliases;  /* aliases */
            int s-port;        /* port (network-byte order) */
            char *s_proto;     /* protocol */ };
```

Example: struct servent *pserv;

```
        pserv=getservbyname("ftp","tcp"); /* FTP using TCP */
```

DNS| getservbyport()

#include <netdb.h>

struct **servent** ***getservbyport** (int **port,** const char ***protoname**);

- Searches for a service that matches the specified *port* and *protocol* (optional)
- Return value: a pointer to *struct servent* on success, NULL on error

Obs.: *port* must be specified in *network byte order*

Example:

```
struct servent *pserv;
pserv=getservbyport( htons(53), ”udp”); /* DNS using UDP */
pserv=getservbyport( htons(21),”tcp”); /* FTP using TCP */
```

DNS| getaddrinfo()

```
#include <netdb.h>
int getaddrinfo (
    const char *hostname,
    const char *service,
    const struct addrinfo *hints,
    struct addrinfo **result ) ;
```

Hostname or an IPv4/IPv6 address as string

Service's port or name ("http","pop",..) (see /etc/services file)

Specifies criteria for refining the return value

- Obs. *hostname*, *service*, *hints* – input parameters
- Return value: 0 on success, !=0 on error
- Recommended to be used for IPv4 and IPv6
- Combines functionalities of: gethostbyname(), getservbyname(), getservbyport()

DNS| getaddrinfo()

```
struct addrinfo {
    int ai_flags;         /* AI_PASSIVE, AI_CANONNAME */
    int ai_family;        /* AF_INET, AF_INET6, AF_UNSPEC */
    int ai_socktype;      /* SOCK_STREAM or SOCK_DGRAM */
    int ai_protocol;      /* 0 (auto) or IPPROTO_TCP, IPPROTO_UDP */
    socklen_t ai_addrlen; /* ai_addr length */
    char *ai_canonname; /* host's canonical name */
    struct sockaddr *ai_addr; /* socket's binary address */
    struct addrinfo *ai_next; /* pointer to the next structure from
    the list */
};
```

DNS| getaddrinfo()

Discussion:

- If the function successfully returns, **result** will point to a list of **struct addrinfo** elements

  Cases when we can have multiple structures:
  - If a hostname has multiple IP addresses, then a structure is returned for each one
  - If the service is offered for different types of *sockets*, then a structure in returned for each type of *socket*
- The data returned by **getaddrinfo**() in **struct addrinfo** ****result** can be used like this:
  - for **socket**() : **ai_family**, **ai_socktype**, **ai_protocol**
  - for **connect**() or **bind**(): **ai_addr** and **ai_addrlen**

- **freeaddrinfo()**

DNS| getnameinfo()

```
#include <netdb.h>
int getnameinfo (
    const struct sockaddr *sockaddr,
    socklen_t addrlen,
    char *host,
    socklen_t hostlen,
    char *serv,
    socklen_t servlen,
     int flags) ;
```

socket's address

name of the returned host

the name of the service

NI_NOFQDN -> **host** will contain only the name of the host, not the full domain name

- Replaces gethostbyaddr() and getservbyport()
- Returns: 0 on success, !=0 on error

DNS| IDN

- **Internationalized Domain Names (IDN)**
  - Extension which enables the use of Unicode characters for domain names, not only ASCII ones

    https://www.icann.org/resources/pages/idn-2012-02-25-en

    16 November 2009 – Registration of ccIDN (or IDN ccTLD domains)

    > 2010-01: ICANN announces that Egypt, the Russian Federation, Saudi Arabia, and the United Arab Emirates were the first countries to have passed the Fast Track String Evaluation within the IDN ccTLD domain application process.

  - Can be used for *phishing attacks* (… details in a future lecture)

DNS| administration

- DNS root is officially administered by Internet Corporation for Assigned Names and Numbers (ICANN)

- There are other organizations that offer alternative roots, like OpenNIC (Network Information Center), Handshake, or Yeti DNS

Summary

- Domain Name System (DNS)
  - Characterization
  - Organization
  - Configuration
  - Commands, Primitives
  - Internationalized Domain Name (IDN)

Questions?